



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 31.07.2026

In addition to part I (General Handout for all courses appended to the Timetable) this portion gives further specific details regarding the course.

Course No : MPBA G517
Course Title : Big Data Analytics
Instructor-in-Charge : Praveen Goyal

1. Course Description:

The course imparts skills on big data in analytics using Apache Spark. At the end of the course, students are expected to apply concepts of big data and big data architecture on huge volumes of real time data, and develop product or solutions development.

2. Scope and Objective of the Course:

Apache Spark is the tool taught for handling the concepts of big data. All the illustrations are done on native Spark, and with Python API for Spark i.e., PySpark. Students may use R Spark API for submission of assignments.

3. Text Books: Learning Spark: Lightning-Fast Data Analytics, Second Edition by Jules S. Damji et al., (2020)

4. Reference: Nil

5. Course Plan:

Lecture Hours	Lecture Module	Reference	Learning outcomes
1	Introduction to Big Data & Evolution of Tools to Handle Big Data	Class Notes	To analyse the reasons for big data architecture, introduce different tools to handle big data.
2-3	Handling Big Data in the Cloud (AWS)	Class Notes	To understand how big data architecture is handled and large volumes of data are managed on the cloud (AWS).
4	Handling Big Data in the Cloud (Google Cloud)	Class Notes	To understand how big data architecture is handled and large volumes of data are managed on the cloud (Google Cloud or Azure).
5-6	Apache Spark: For Big Data Processing (Session-I)	Chapter 1 Introduction to Apache Spark: A	To understand how Apache Spark is useful for processing of big data.
7	Spark Basics & Resilient Distributed Datasets (RDDs)	Chapter 3 Apache Spark's Structured APIs	To familiarize with Spark ecosystem (PyShell, Spark UI, Spark Application & Spark Session).
8	Spark SQL, Dataframes & Datasets (Session-I)	Chapter 3 Apache Spark's Structured APIs	To apply and perform operations on data frame and dataset on Spark



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9	Spark SQL Based solution development	Chapter 3 Apache Spark's Structured APIs	To apply and perform operations on data frame and dataset on Spark
10	Monitoring & Troubleshooting Spark Jobs	Class Notes	To create jobs in Spark, monitor and troubleshoot jobs in Spark.
11	Real-time streaming using Apache Spark	Class Notes	To perform real-time streaming using Spark
12	Application of Machine Learning Algorithms on Big Datasets Through Spark (Session-I)	Chapter 10 Machine Learning with MLlib & Class Notes	To apply machine learning algorithms on big datasets in Spark.
13-18	AWS Cloud based NLP-LLM solution development including SageMaker	Chapter 10 Machine Learning with MLlib & Class Notes	To apply machine learning algorithms on big datasets and create pipelines.
19-20	Containerization	Class Notes	To understand concept of container and docker
21-23	Cloud based solution architecture development	Class Notes	Details around different components to use and evaluate the data flow based upon different use cases
24-26	Airflow	Class Notes	Understanding the development of acyclic workflows to develop production grade solutions
27-30	MLops + Tools overview	Class Notes	Understanding the life cycle of MLops and working with tools
Practical Sessions-wise Plan:			
1	Apache Spark Installation & integrate with Python using PySpark API 2. Explore Spark ecosystem (PyShell, Spark UI, Spark Application & Spark Session)	Chapter 2 Downloading Apache Spark & Getting Started	To familiarize with Spark ecosystem (PyShell, Spark UI, Spark Application & Spark Session)
2	The DataFrame & Dataset API	Chapter 3 Apache Spark's Structured APIs	To apply and perform operations on data frame and dataset on Spark.
3	Built-in data sources	Chapter 4	To create data frames from multiple data sources
4	Perform SQL operations in Spark	Chapter 5 Spark SQL and Dataframes: Interacting with External Data Sources	To create data frame by reading data from SQL database, and perform SQL operations in Spark
5	Monitoring & Troubleshooting Spark Jobs	class notes	To create jobs in Spark, monitor and troubleshoot jobs in Spark.



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6	Real-time streaming using Apache Spark	class notes	To perform real-time streaming using Spark
7	Application of Machine Learning Algorithms on Big Datasets Through Spark (Session-I)	Chapter 10 Machine Learning with MLib & Class Notes	To apply machine learning algorithms on big datasets in Spark.
8	Application of Machine Learning Algorithms on Big Datasets Through Spark (Session-II)	Chapter 10 Machine Learning with MLib & Class Notes	To apply machine learning algorithms on big datasets in Spark.
9	Application of Deep Learning Algorithms on Big Datasets Through Spark (Session-III)	Chapter 10 Machine Learning with MLib & Class Notes	To apply machine learning algorithms on big datasets in Spark.
10	Application of Deep Learning Algorithms on Big Datasets Through Spark (Session-IV)	Chapter 10 Machine Learning with MLib & Class Notes	To apply machine learning algorithms on big datasets in Spark.

6. Evaluation Scheme: Evaluation components assess a range of learning outcomes: basic comprehension of the topics taught, application of different assignments in the lab skills learnt, and ability to solve big data problems. The proposed evaluation scheme is given below.

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/Open Book)
Lab Evaluation		5		
Class participation		5		
Mid-semester evaluation	1.5 h	25	09/10 AN2	TBA
Mid-semester Group Project		15		
End-semester Group Project		15		
Comprehensive Examination	3 h	35	14/12 AN	TBA

7. Chamber Consultation Hour: Will be informed during class.

8. Notices: Notices will be shared in Nalanda or Google Classroom

9. Make-up Policy: Genuine cases such as hospitalization (upon production of the relevant documents/proof) will be considered upon discussion with individuals/team for any make ups for the final Exam.

10. Note (if any):

Praveen Goyal
Instructor-in-charge
Course No. MPBA G517